Notebook: all_algo_comparsions/logistic_regression_comparison.ipynb

[--- MARKDOWN CELL ---]

Logistic Regression - Algorithm Comparison

Comparing from-scratch vs sklearn on Breast Cancer and Wine datasets.

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```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_breast_cancer, load_wine
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression as SklearnLR
from sklearn.metrics import accuracy_score, confusion_matrix
np.random.seed(42)
```

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```
class LogisticRegressionScratch:
    def __init__(self, lr=0.01, epochs=1000):
        self.lr, self.epochs = lr, epochs
        self.w, self.b, self.cost_history = None, None, []
    def sigmoid(self, z):
        return 1 / (1 + np.exp(-np.clip(z, -500, 500)))
    def fit(self, X, y):
        m, n = X.shape
        self.w, self.b = np.zeros(n), 0
        for _ in range(self.epochs):
            pred = self.sigmoid(X @ self.w + self.b)
            self.w -= self.lr * (X.T @ (pred - y)) / m
            self.b -= self.lr * np.sum(pred - y) / m
        return self
    def predict(self, X):
        return (self.sigmoid(X @ self.w + self.b) >= 0.5).astype(int)
    def score(self, X, y):
        return accuracy_score(y, self.predict(X))
```

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```
# Breast Cancer
bc = load_breast_cancer()
X_train, X_test, y_train, y_test = train_test_split(bc.data, bc.target, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train_s, X_test_s = scaler.fit_transform(X_train), scaler.transform(X_test)
lr_scratch = LogisticRegressionScratch(lr=0.1, epochs=1000).fit(X_train_s, y_train)
lr_sklearn = SklearnLR(max_iter=1000).fit(X_train_s, y_train)
print(f"Breast Cancer - Scratch: {lr_scratch.score(X_test_s, y_test):.4f}, Sklearn: {lr_sklearn.score(X_test_s,
y_test):.4f}")
```

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```
# Wine (Binary)
wine = load_wine()
y_bin = (wine.target != 0).astype(int)
X_train, X_test, y_train, y_test = train_test_split(wine.data, y_bin, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train_s, X_test_s = scaler.fit_transform(X_train), scaler.transform(X_test)
lr_scratch = LogisticRegressionScratch(lr=0.1, epochs=1000).fit(X_train_s, y_train)
```

```
lr_sklearn = SklearnLR(max_iter=1000).fit(X_train_s, y_train)
print(f"Wine - Scratch: {lr_scratch.score(X_test_s, y_test):.4f}, Sklearn: {lr_sklearn.score(X_test_s, y_test):.4f}")
```

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```
# Comparison plot
datasets = ['Breast Cancer', 'Wine']
scratch = [0.96, 0.97]
sklearn = [0.97, 0.97]
x = np.arange(2)
plt.figure(figsize=(8,5))
plt.bar(x-0.2, scratch, 0.4, label='From Scratch')
plt.bar(x+0.2, sklearn, 0.4, label='Sklearn')
plt.xticks(x, datasets)
plt.ylabel('Accuracy')
plt.title('Logistic Regression Comparison')
plt.legend()
plt.ylim(0.9, 1.0)
plt.show()
```